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# Natural deep eutectic solvents efficient catalytic conversion of cellulose to total reducing sugars (TRS)

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ABSTRACT

Lignocellulosic biomass contains about 45% cellulose. The total reducing sugars (TRS) obtained from lignocellulosic materials is beneficial to the sustainable development. Natural deep eutectic solvents (NADESs) provide an attractive alternative for the "green" conversion of cellulose. In this work, series of amino acids based NADESs with $FeCl_3 6H_2O$ as ligands catalyst were synthesized and characterized by in situ IR, FT-IR, $^1H$ NMR and TG-DTA. NADESs showed great thermal stability. The efficient hydrolysis performance of cellulose in NADESs was investigated under mild conditions. The influencing factors such as the type of NADESs, reaction temperature, reaction time, and $FeCl_3 6H_2O$ addition on the TRS yield were studied. The optimized process conditions for hydrolyzing cellulose in L-Proline/ Ethylene glycol/$FeCl_3 6H_2O$ were molar ratio 1: 2: 1, reaction time 120 min and temperature 120 °C. The yield of total reducing sugar was 66.7%. The conversion mechanism was also investigated.

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## 1. Introduction

Concerns about fossil carbon depletion and global climate change have led to an increasing focus on biomass conversion to sustainably supply liquid fuels and valuable chemicals. Therefore, the search for renewable energy to replace fossil fuels has become a top priority. According to the forecast made by the US Department of Energy, the amount of liquid fuels consumed in the US from biofuels should increase 200% due to the Federal Renewable Fuels Standard, which requires 36 billion gallons of renewable fuel to be blended into transportation fuels by 2022 [1]. Recently, biomass conversion has become a research hotspot of many scholars at home and abroad, and different methods have been developed to convert biomass into fuels, chemicals and materials [2–5].

Biomass resources are the only renewable organic carbon resources in nature. Lignocellulose is the most abundant green renewable resource and the main structural form of plants [6–9]. Lignocellulose consists of cellulose, hemicellulose and lignin, with an annual production of about 170 billion tons, of which about 45% is cellulose [10–12]. Cellulose is the most prolific component of lignocellulose, which is a linear polymer compound obtained by linking D-glucopyranose to 1,4-β glycoside bonds, and the chemical formula of cellulose is $(C_6H_{10}O_5)n$ [13,14]. The cellulose molecule contains a large amount of hydroxyl groups inside, and since hydrogen bonds are easily generated between the hydroxyl group and the hydroxyl group, a large number of hydrogen bonds exist inside and between the cellulose molecules, forming a strong hydrogen bond network structure [15–17]. Therefore, cellulose is extremely difficult to dissolve in water or other conventional organic solvents, which seriously affects the contact between the catalyst and cellulose molecules and becomes the biggest obstacle to cellulose conversion.

In the past few decades, research on the conversion of cellulose has developed rapidly. Conversion of cellulose into various value-added chemicals and fuels, such as various alcohols [18], gluconic acid [19], 5-hydroxymethylfurfural (HMF) [20], lactic acid [21], pentanoic acid esters [22], 2,5-dimethylfuran [23], and levulinic acid [24], has been studied extensively in recent years. The catalytic oxidation of cellulose to total reducing sugars (TRS) is one of the most promising pathways. TRS is used as a fine chemical product, and it is widely used in the pharmaceutical, food and industrial industries.

In recent decades, ionic liquids (ILs) have been thought of as suitable solvents instead of sulfuric acid for conversion owing to their unique properties, including negligible vapor pressure, high thermal stability, and good solvent power for different kinds of substances [25]. In 2018, Saliha Saher [26] synthesized a pyridinium-based ionic liquid $[C_4C_1Py][Cl]$, which was used for the dissolution of cellulose, with a solubility of up to 28% at 110 °C, and continued in the presence of the metal salt

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$CoCl_2$ as a catalyst. The cellulose was converted to a TRS yield of 78% at 4 h. However, high prices, poor toxicity and biodegradability limit the further application of traditional IL in the field of green chemistry.

As an attractive alternative to ILs and conventional organic solvents, deep eutectic solvents (DESs) have emerged and have evolved significantly [27]. DESs are usually binary mixtures of hydrogen bond acceptors (HBA) and hydrogen bond donors (HBD). At a certain molar ratio, the freezing point is lower than the melting point of the pure compound constituting DESs [28]. DESs have similar solvent properties to ILs such as the low-cost, good renewability and low toxicity [29]. Moreover, they are easy to prepare under mild conditions.

In recent works, there are many DESs which have been prepared for cellulose conversion. In 2012, Francisco et al. [30]. obtained a variety of DES solvent systems using representative natural substances commonly found in nature such as amino acids, choline chloride, niacin, malic acid and lactic acid. It was found that most DES have weak cellulose conversion ability, and DES composed of proline and malic acid can also convert a certain amount of cellulose. In 2015, Procentese [31] synthesized glycerol/choline chloride (Gly/ChCl), urea/choline chloride (Urea/ChCl) and imidazole/choline chloride (Im/ChCl) DESs for corn core conversion. In this process, it was found that the DES promotes the saccharification conversion process by removing lignin and reducing the crystallinity of the cellulose, and a total of 41 g of fermentable sugar (27 g of glucose and 14 g of xylose) can be recovered from 100 g of corn cob, accounting for 76% of the initial available reducing sugar. In 2016, Wahlstrom et al. [32] synthesized three choline chlorides: boric acid (5:2 M ratio), choline chloride: glycerol (molar ratio 1:1), and betaine: glycerol (molar ratio 1:1). The composition of the DES converted the wheat straw, and the stability of the wheat straw to TRS, and the yield after the residue hydrolysis were 33.0%, 20.0% and 19.0%, respectively. Overall, the conversion of cellulose in DESs is not ideal. Therefore, it is important to find efficient DESs with high conversion ability and selectivity to biomass.

As a kind of Lewis acid, Iron (III) chloride hexahydrate ($FeCl_36H_2O$) is widely used as catalyst in chemical reactions such as esterification, condensation, etherification and oxidation. It has the advantages of high catalytic activity, reusability and low price [33]. $FeCl_36H_2O$ catalyzed hydrolysis reaction conditions are mild, less corrosion to equipment, simple operation, short reaction cycle, low cost, and environmental protection.

In this research, it was devoted to the synthesis of a novel series of amino acid based natural deep eutectic solvents (NADES) with $FeCl_36H_2O$ as ligands catalyst, and utilized to convert cellulose into the total reducing sugar. The physical properties of NADESs include freezing point, solvation parameters, pH, thermal stability and synthesis processes were investigated. The chemical properties of the NADESs such as $^1H$ NMR and FT-IR were also measured. The effects of different external factors, such as different sorts of NADESs, temperature, time and $FeCl_36H_2O$ addition on cellulose conversion were studied. The conversion mechanism was also developed.

## 2. Experiments

### 2.1. Materials

The cellulose was cotton linter pulp with a polymerization degree of 575.6 (Helon, China). The cotton linter pulp was cut into small pieces. After being dried at 100 °C for 12 h, the cellulose was prepared and used in the present study. $N_2$ (99.99%) was obtained from Beijing Haipu Gases Co., Ltd., Beijing, China. L-Proline, L-Lysine and L-Glutamic acid (99%) were purchased from Huayang Chemical Co., Ltd., Hebei, China. Iron(III) chloride hexahydrate (99%), Methanesulfonic acid (MSA) and *p*-Toluenesulfonic acid (TsOH) were obtained from Shanghai Aladdin Biochemical Technology Co., Ltd. Ethylene glycol and Glycerol (99%) was obtained from Tianjin Yongda Reagent Co., Ltd., Tianjin, China. All chemical reagents used in this work were of analytical grade purity and were used directly.

### 2.2. Synthesis of NADESs

A series of natural deep eutectic solvents were synthesized according to the previously methods [34]. The amino acids (L-Proline, L-Lysine, L-Glutamic acid, etc.) were used as hydrogen bond acceptors, ethylene glycol, *p*-toluenesulfonic acid, Methanesulfonic acid, etc. were used as hydrogen bond donors, and $FeCl_36H_2O$ as ligands were added to a 100 mL round bottom three-necked flask at a certain ratio. The mixture was then heated at a heating rate of 5 °C·h$^{-1}$ from room temperature to 100 °C under magnetic stirring. The container was also protected with nitrogen until a uniform, stable liquid formed. The sample was then placed in a forced air oven at 60 °C for 24 h, and the products were obtained (Fig. 1).

### 2.3. Conversion of cellulose in NADESs

5 g of prepared NADESs was added to a 100 mL round bottom flask, and then heated at a heating rate of 5 °C·h$^{-1}$ from room temperature to 140 °C under magnetic stirring. Then 1 g cellulose was added to the flask and allowed to react for 2 h at a set temperature from 90 °C to 140 °C. The reaction was terminated immediately by placing in an ice water bath.

The cellulose conversion rate is calculated by measuring the change in mass before and after the reaction. After the reaction, the mixed solution was rapidly cooled, centrifuged and filtered. Ethanol was added to the solid residue as back extractant of cellulose. The insoluble cellulose solids were dried and weighed to calculate the conversion of cellulose [35].

$$\text{Cellulose conversion} = \left(1-\frac{W_R}{W_D}\right) \times 100\%$$

In the formula, $W_R$——mass of residue, g. $W_D$—mass of cellulose, g.

In the regeneration of the NADES, it followed three steps. Firstly, the unreacted cellulose solid was filtered when the reaction stopped. Then, the upper solution was precipitated with ethanol at a volume fraction of 80% (volume ratio of solution to ethanol was 1:4) [36–40]. After the solution stood for 4 h, and filtered the insoluble crude reducing sugars, and rotary evaporated the ethanol, NADESs was regenerated for the recycling. The crude reducing sugars sample was then extracted by 100 mL butanone. The left L-Proline was regenerated being reused.




**Fig. 1.** Photographs of a series of NADESs. Solutions labeled A, B, C, D, and E correspond to L-Pro/EG, L-Lys/EG, L-Pro/TsOH, L-Glu/Gly, L-Pro/MSA.

### 2.4. Analytical methods

Nuclear magnetic spectra ($^1$H NMR) were recorded using a nuclear magnetic resonance spectrometer (Bruker, Avance 500 MHz, Germany) at room temperature. The solvent was $D_2O$. The Fourier transform infrared spectra was recorded on a Fourier transform infrared spectrometer (FTIR) (Bruker Tensor 27, Germany) with wavenumbers from 4000 to 400 $cm^{-1}$. The in situ infrared spectrometer system (Mettler Toledo, React IR 15, Switzerland) can be used for tracing and detecting the chemical reactions over the entire course of the experiment in order to analyze the mechanism and process of the reaction. Thermogravimetric (TG) (Netzsch, STA449 F5, Germany) analysis was performed under helium. The samples for TG analysis were placed in a small crucible with a cover under the helium atmosphere with a flow rate of 20 mL/min and heated at a temperature range of 50 °C ~ 560 °C with a heating rate of 20 °C/min. The pH of the sample was determined by a Seven Excellence multiparameter tester from Mettler Toledo. The freezing point of NADES were determined by DSC 214 (Netzsch, German), the gas atmosphere was $N_2$ with 40.0 mL/min/$N_2$, and the temperature was from −35 °C to 35 °C with a heating rate of 10 °C/min. The total reducing sugar yield is collected with a UV-6100S UV–vis spectrophotometer (Metash instrument Co., Ltd., China). The scan range is 200 to 500 nm with the interval of 0.5 nm. The monosaccharide is determined with an Agilent1260II high-performance liquid chromatography (HPLC) (Agilent Technologies, USA) with an UV detector and a C-18 column at 30 °C. The eluent is pure water at a flow rate of 0.7 mL/min. The other by-products are determined by a 7890A/5975 gas chromatography–mass spectrometry (GC–MS) (Agilent Technologies, USA) with HP-5MS column (30 m × 0.25 μm, 0.25 mm). The column temperature is from 30 °C to 270 °C, and heating rate of 15 °C/min. The residual solid products of cellulose hydrolysis are analyzed by FTIR, TG, and SEM (S-4800, Hitachi, Japan) analysis.

The polarity scale ($E_T(30)$), dipolarity/polarizability effects ($\pi^*$), hydrogen-bond donating acidity ($\alpha$), and hydrogen-bond basicity ($\beta$) were calculated by Eq. (1) to (6) according to Kamlet-Taft empirical polarity scales:

$$E_T(30) = 28,592/\lambda_{max(RD)} \quad (1)$$

$$\alpha = 0.0649\,E_T\,(30) - 2.03 - (0.72\,\pi*) \quad (2)$$

$$\pi* = 0.649 - 0.314\nu_{(NA)} \quad (3)$$

$$\beta = (1.035\,\nu_{(DENA)} + 2.64 - \nu_{(NA)})/2.80 \quad (4)$$

where,

$$\nu_{(NA)} = 1/\left(\lambda_{max(NA)} \times 10^{-4}\right) \quad (5)$$

$$\nu_{(DENA)} = 1/\left(\lambda_{max(DENA)} \times 10^{-4}\right) \quad (6)$$

Here, $\lambda_{max}$ represents wavelength of maximum absorption.

### 2.5. Calculation of the total reducing sugar yield

After the reaction was completed, a 30–50 mg sample was removed from the reaction vial, and diluted 50 times with distilled water. It was then subjected to DNS assay using a UV–vis spectrophotometer (UV-6100S, Metash instrument Co., Ltd., China) for calculation of total reducing sugar.

The total reducing sugar content was measured by the DNS method, and the reducing sugar content was obtained by a standard curve. The total reducing sugar yield was calculated according to formula (7).

**Table 1**
The composition of NADES at the eutectic point.

| DES | Optimized molar ratio | $T_f$ (°C) | $T_m$ (°C) | $\Delta$(°C) |
|---|---|---|---|---|
| L-Pro/EG | 1: 2 | 24.6 | 230 | 205.4 |
| L-Lys/EG | 1: 6 | 26.4 | 215 | 188.6 |
| L-Pro/TsOH | 2: 1 | 29.5 | 230 | 200.5 |
| L-Glu/Gly | 1: 8 | 32.8 | 160 | 127.2 |
| L-Pro/MSA | 1: 2 | 38.5 | 230 | 191.5 |

$$Y_{TRS} = \frac{\omega_{TRS}}{W_D \times C_C} \times 0.9 \times 100\% \quad (7)$$

In the formula, $Y_{TRS}$:—yield of total reducing sugar, %. $\omega_{TRS}$—mass of total reducing sugar, g. $C_C$—carbohydrate content in sample. $W_D$—mass of raw cellulose, g.

## 3. Results and discussion

### 3.1. Synthesis of natural deep eutectic solvents

In the synthesis of NADESs, the freezing point ($T_f$) was used to represent the eutectic of the mixture, indicating an interaction occurred between the mixtures to form a new liquid material [41–43]. It is a typically character of NADESs. From the results of Table 1, it could be seen that different amino acids and different hydrogen bond donors can form a variety of deep eutectic solvents. The $T_f$ of L-Pro/EG, L-Lys/EG, L-Pro/TsOH, L-Glu/Gly and L-Pro/MSA under optimized ratio were 24.6 °C (1: 2) and 26.4 °C (1: 6) 29.5 °C (2: 1), 32.8 °C (1: 8) and 38.5 °C (1: 2), respectively (Figs. S1–S5). The considerably difference between the freezing point of the deep eutectic system and that of raw material was from 127.2 °C to 205.4 °C. The significant depression indicated some chemical reaction occurred between the amino acid and the hydrogen bond donor, and the NADES were prepared successfully [44].

Label 1:$T_f$ is the freezing point of the deep eutectic solvents; $T_m$ is melting point of HBA raw materials; $\Delta$ is the difference between $T_f$ and $T_m$.

### 3.2. Properties of NADESs

#### 3.2.1. Thermostability

Thermal stability is a significant parameter related to cellulose conversion in NADESs. The thermodynamic properties of the synthesized NADESs were analyzed by DTG-TGA analysis (Fig. 2). The temperatures at different ratios of mass loss ($T_{10\%}$ and $T_{50\%}$) and the temperature of the maximum loss mass rate ($T_{max}$) were listed in Table 2 [45]. As can be seen from the figure, the weight loss of L-Pro/EG, L-Lys/EG, L-Pro/TsOH, L-Glu/Gly and L-Pro/MSA is 10% at 138.9 °C, 131.2 °C, 258 °C, 201.4 °C and 289 °C, respectively. It indicates that the prepared NADESs have good thermal stability to meet the hydrolysis of cellulose.

The synthesis process of the NADESs was monitored with FT-IR and in situ IR spectroscopy, respectively (Fig. 3(a–d)). In Fig. 3(a), the peak at a wavelength of 1621 $cm^{-1}$ denotes the N—H stretching vibration of the L-Proline. The characteristic peak 3355 $cm^{-1}$ is attributed to the -OH group in ethylene glycol. Comparing with the spectra of raw materials (L-Proline and ethylene glycol), the redshifts and increased peak breadth occur. The C—O stretching characteristic peak appears at 1039 $cm^{-1}$ for pure ethylene glycol, and it shifts to 1041 $cm^{-1}$ in the forming of NADESs between L-Proline and ethylene glycol. In the fingerprint region, some changes also occur alongside. All the changes in the spectrum demonstrate that L-Proline and ethylene glycol are interconnected through hydrogen bonds. The FT-IR results of other NADESs are referred to Figs. S6–S9.

To verify the reaction between L-Proline and ethylene glycol, $^1$H NMR results were shown in Fig. 4. It can be observed that the shift

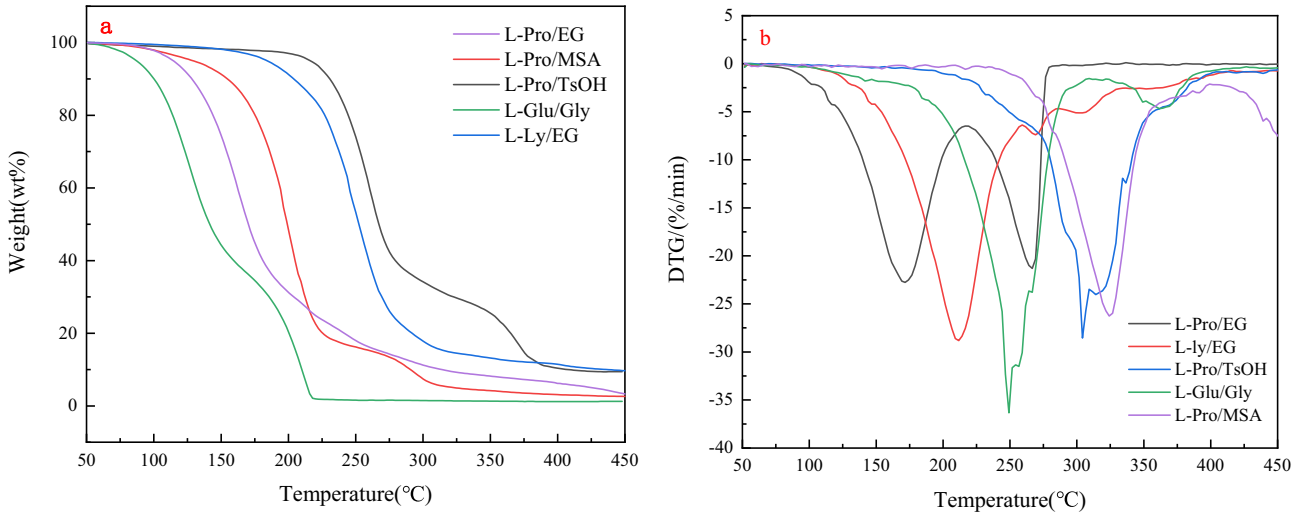



**Fig. 2.** (a) TG (in nitrogen) curves of NADESs; (b) DTA (in nitrogen) curves of NADESs.

**Table 2**
Thermal properties of NADESs.

| NADESs | $T_{10\%}$ | $T_{50\%}$ | $T_{max}$ |
|---|---|---|---|
| L-Pro/EG | 138.9 | 187.5 | 171 |
| L-Lys/EG | 131.2 | 170.2 | 162 |
| L-Pro/TsOH | 258 | 314.4 | 304.6 |
| L-Glu/Gly | 201.4 | 251.4 | 247 |
| L-Pro/MSA | 289 | 332 | 325 |

and strength of the typical peak at 3.67 ppm is not greatly changed. However, the peak of $CH_2$— in L-Proline moves decline from 3.35 to 3.33 ppm. The peaks at 2.33 and 3.40 ppm are assigned to —$CH_2C$=O on L-Proline and —$CH_2N$— on L-Proline, respectively. These peaks all show slight changes in the height and strength, and the results were consistent with that of in situ IR spectra. The $^1$H NMR results of other NADESs are referred to Figs. S10–S13.

The impact of $FeCl_3 \cdot 6H_2O$ on NADESs was investigated, and the results were shown in Fig. 5. In the low-wavenumber region, there is no

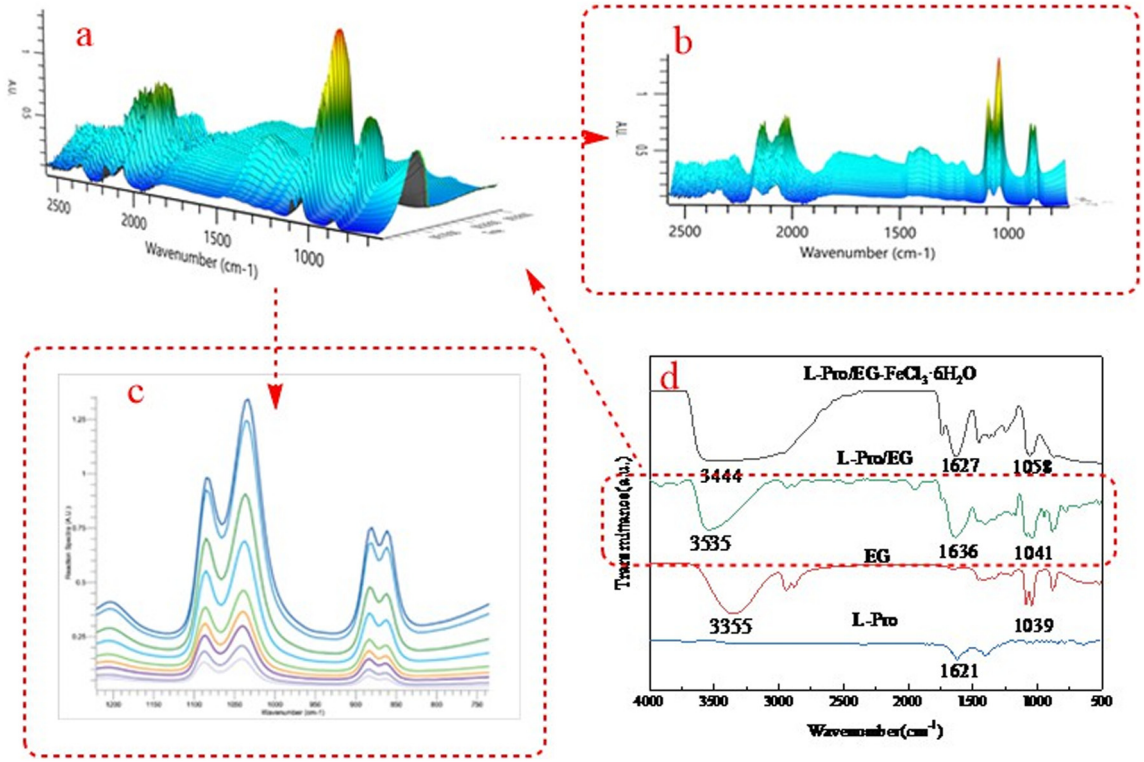



**Fig. 3.** FT-IR and in situ IR spectra of L-Pro/EG NADESs (molar ratio of 1: 2). (a): In situ IR spectra of L-Pro/EG NADESs (1:2); (b): A front view of an a; (c): A partial enlarged view of (a) ranging from 800 to 1200 $cm^{-1}$; (d): FT-IR of L-Pro/EG/$FeCl_3 \cdot 6H_2O$ and raw materials (L-Pro/EG, $FeCl_3 \cdot 6H_2O$).

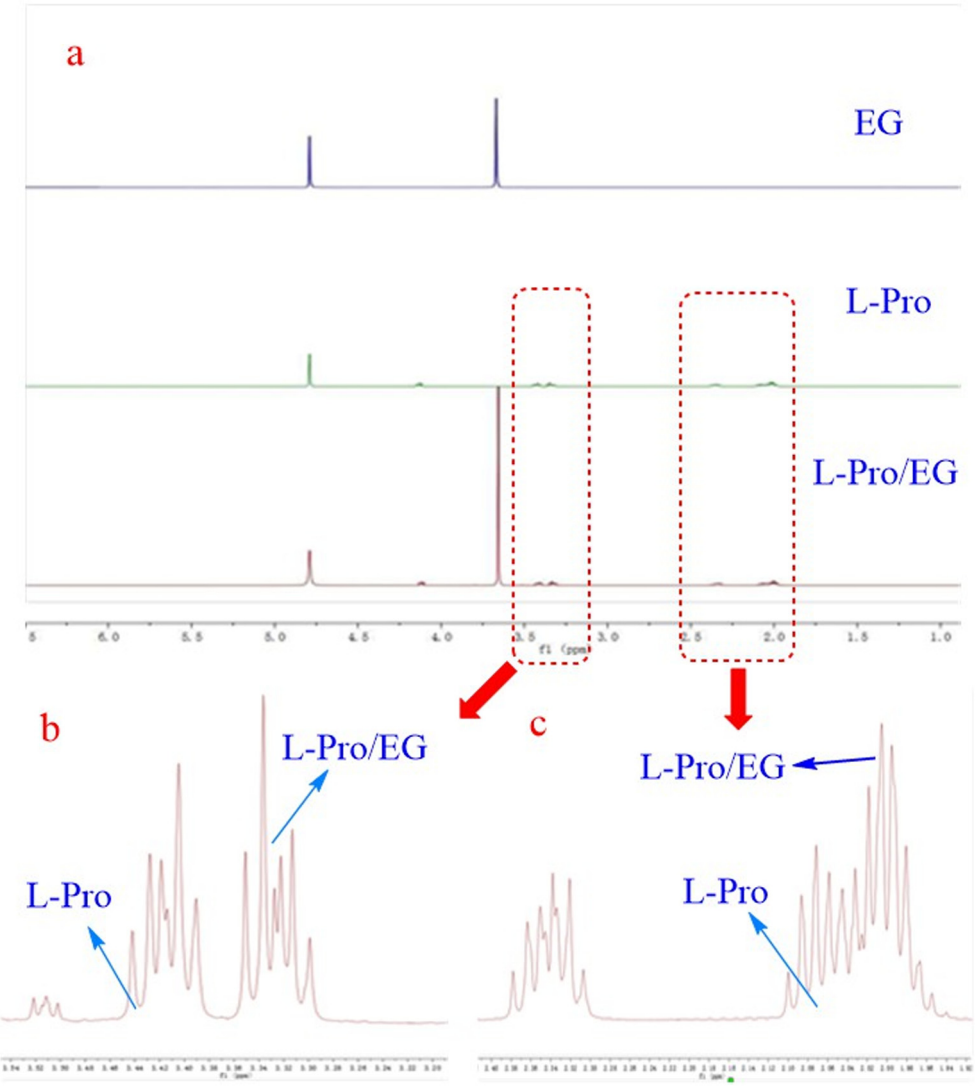



**Fig. 4.** (a) $^{1}$H NMR spectra of the NADESs (L-Pro/EG); (b) a partial enlarged view of (a) ranging from 3.54 to 3.20 ppm; (c) a partial enlarged view of (a) ranging from 2.40 to 2.92 ppm.

obvious new peak appeared. However, the intensity of the peak decreases in the range of 1000 to 1700 $cm^{-1}$. Especially, the decreased peak at 1630 $cm^{-1}$, attributed to the vibration of C—O on RCOOH, demonstrates that the L-Pro/EG NADESs is decomposed to a certain extent. The corresponding change in the intensity of the infrared peaks in Fig. 5a confirms the formation of NADESs into NADESs/$FeCl_3 \cdot 6H_2O$. The C—O stretching characteristic peak (1041 $cm^{-1}$) in L-Pro/EG NADESs shows a red shift to 1627 $cm^{-1}$ in L-Pro/EG/$FeCl_3 \cdot 6H_2O$. Evermore, the stretching peak of N—H at 3535 $cm^{-1}$ in L-Pro/EG NADES shifted to 3444 $cm^{-1}$ in L-Pro/EG/$FeCl_3 \cdot 6H_2O$.

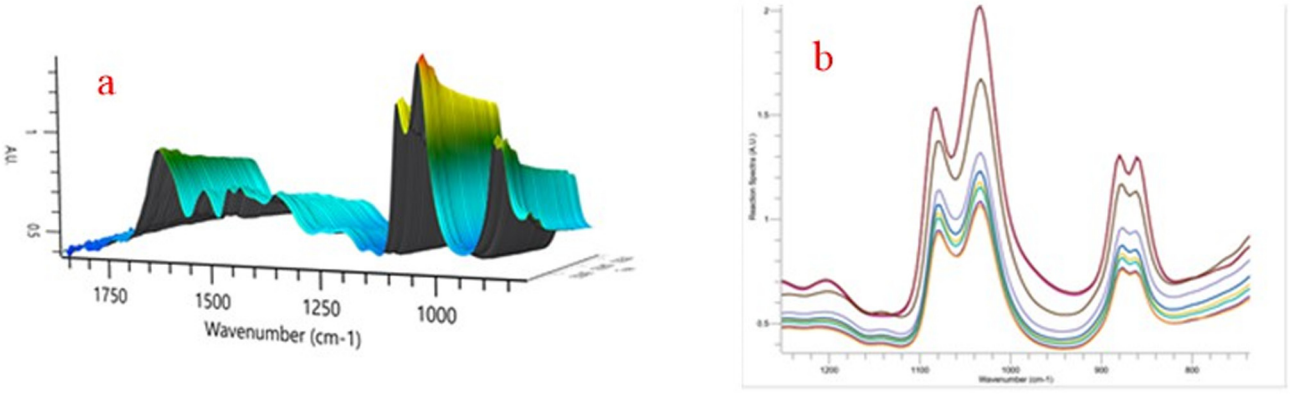



**Fig. 5.** (a): In situ IR spectra of L-Pro/EG/$FeCl_3 6H_2O$ (1:2:1); (b) a partial enlarged view of (a) ranging from 800 to 1800 $cm^{-1}$.

*3.2.2. PH*

The pH reflects the acidity of the solution, and it directly affects the conversion of cellulose. As shown in Fig. 6, the pH of NADESs was about 4–5 in the absence of $FeCl_3 \cdot 6H_2O$, and it showed weak acidity. However, the acidity of the system increased to -1.8- -1.1 when $FeCl_3 \cdot 6H_2O$ was added. L-Lys/EG/$FeCl_3 \cdot 6H_2O$ was the most acidic solvent, but the yield of TRS was not the highest. It was possible that higher acidity would enhance certain side reactions, such as dehydration of glucose to 5-hydroxymethylfurfural. The acidity of L-Pro/EG/$FeCl_3 \cdot 6H_2O$ was weaker than that of L-Lys/EG/$FeCl_3 \cdot 6H_2O$, but the yield of TRS was the highest among these five systems. It could be concluded that proper acidity was favorable for increasing the TRS yield.

*3.2.3. Solvation properties*

The Kamlet-Taft solvation model is often used to study the biomass conversion in liquid liquids or DESs. Solvation parameters include hydrogen bond acidity ($\alpha$), hydrogen bond basicity ($\beta$), and polarizable effect ($\pi^*$). The dipolar/polarizability is used to describe the non-specific part of the interaction between the solvent and the solute Van der Waals, such as dispersion, induction, and electrostatic interactions. Hydrogen bond basicity characterizes a compound's ability to give an electron pair as a hydrogen bond donor, and hydrogen bond acidic alpha indicates the compound's ability to accept an electron pair as a hydrogen bond acceptor.

As shown in Table 3, L-Pro/EG NADESs differed greatly from the other four types of NADESs in solvation properties, which had higher $\alpha$ values (1.58), lower $\beta$ values (0.53), smaller $\pi^*$ (0.73), and $E_T$ (30) (68.27 kcal $mol^{-1}$). These solvation parameters insured that the hydrogen-bond interaction between L-Pro and EG was stronger than others. Besides, the L-Pro/EG NADESs solution had weaker polarizability and stronger polarity. It indicated that it would have a better ability to bind electrons, which was beneficial for the formation of hydrogen bonds between NADESs and cellulose, thereby promoting the conversion of cellulose.

*3.2.4. Moisture content*

The moisture contents of different NADESs/$FeCl_3 6H_2O$ were measured (Fig. 7). The moisture content of L-Pro/EG, L-Lys/EG, L-Pro/TsOH, L-Glu/Gly, and L-Pro/MSA NADESs were form 1.32 wt% to 1.73 wt%. L-Pro/EG was much hydrophilic than other NADESs with the highest moisture content of 1.73 wt%. The reason might be for the presence of a large number of hydroxyl groups in L-Pro. However, the water content of NADESs increased significantly after the catalyst $FeCl_3 \cdot 6H_2O$ was added owning to the crystal water. The water content would interact with the hydrogen and oxygen on the hydroxyl groups of the cellulose, thus affecting the cellulose conversion.

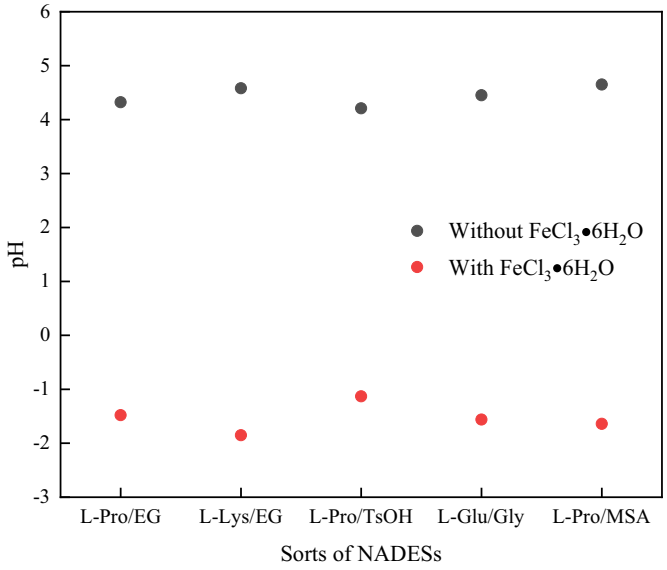



**Fig. 6.** The pH of sorts of NADESs and NADESs/$FeCl_3 \cdot 6H_2O$.

**Table 3**
Kamlet-Taft solvation properties of the natural DESs (25 °C).

| NADESs | $E_T(30)$ (kcal $mol^{-1}$) | $\pi^*$ | $\alpha$ | $\beta$ |
|---|---|---|---|---|
| L-Pro/EG | 68.27 | 0.73 | 1.58 | 0.53 |
| L-Lys/EG | 59.87 | 1.47 | 0.96 | 1.48 |
| L-Pro/TsOH | 38.42 | 1.39 | 0.82 | 1.36 |
| L-Glu/Gly | 27.45 | 1.28 | 0.46 | 1.28 |
| L-Pro/MSA | 52.64 | 1.52 | 0.57 | 1.44 |

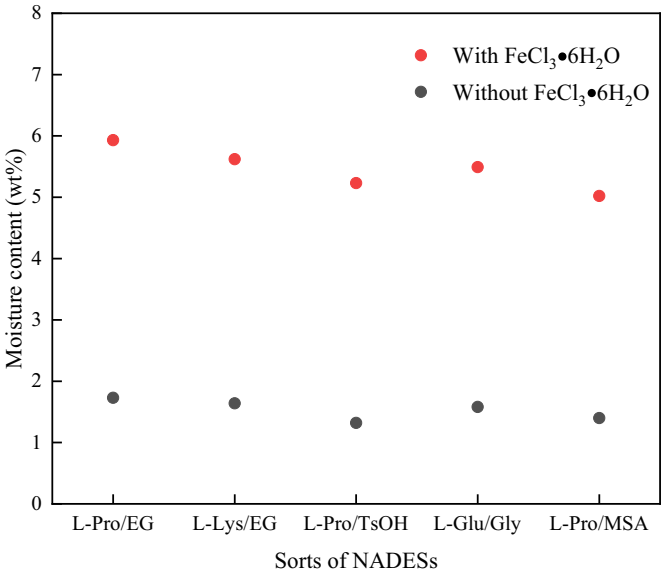



**Fig. 7.** Moisture content of sorts of NADESs.

*3.3. Cellulose conversion in NADESs*

*3.3.1. Effects of NADESs types on cellulose conversion*

It could be seen from Table 3 that, the cellulose conversion yield of TRS was lower catalyzed by NADESs without $FeCl_3 6H_2O$ (Table 4, entries 1–5). Among them, the catalytic performance of L-Pro/EG NADES was slightly better than that of others. It could achieve 20.6% cellulose conversion, and the total reducing sugar yield was 15.2%. The better catalytic activity may be related to the polar structure of L-Pro/EG. After acidic $FeCl_3 6H_2O$ was added, the conversion of cellulose and the yield

**Table 4**
The conversion of cellulose into TRS in different catalyst systems.

| Entry | NADES | Additive | Conversion rate (%) | Yield of TRS (%) |
|---|---|---|---|---|
| 1 | L-Pro/EG | – | 20.6 | 15.2 |
| 2 | L-Pro/MSA | – | 18.5 | 14.2 |
| 3 | L-Pro/TsOH | – | 17.5 | 13.1 |
| 4 | L-Glu/Gly | – | 13.2 | 9.7 |
| 5 | L-Ly/EG | – | 14.8 | 10.4 |
| 6 | L-Pro/EG | $FeCl_3 6H_2O$ | 85.6 | 66.7 |
| 7 | L-Pro/MSA | $FeCl_3 6H_2O$ | 80.5 | 46.2 |
| 8 | L-Pro/TsOH | $FeCl_3 6H_2O$ | 83.6 | 47.6 |
| 9 | L-Glu/Gly | $FeCl_3 6H_2O$ | 78.5 | 42.5 |
| 10 | L-Ly/EG | $FeCl_3 6H_2O$ | 82.6 | 46.5 |

of TRS increased significantly to 85.6% and 66.7%, respectively. The results demonstrated the importance of acidity in the conversion of cellulose (Table 4, entries 6–10).

*3.3.2. Effect of temperature on cellulose conversion*

As shown in Fig. 8, the effect of different reaction temperature on cellulose conversion and reducing sugar yield was investigated in the L-Proline/ethylene glycol/$FeCl_36H_2O$ system. It could be seen that the conversion rate of cellulose and the yield of TRS increased firstly and then decreased as the reaction increasing temperature. When the reaction temperature was 90 °C, the yield of TRS was 25.3%. Continuously increasing the reaction temperature could promote a significant increase in the yield of TRS, and the yield of TRS reached the maximum of 66.7% when the temperature was 120 °C. During the hydrolysis, the increasing temperature can enhance the ionization of NADESs, so the chemical interaction between NADESs and the hydrogen and oxygen on the hydroxyl groups of the cellulose molecules is promoted, and the conversion of cellulose is significantly improved. However, higher temperature above 120 °C would lead to the degradation or repolymerization of carbohydrates, and the yield of reducing sugar decreased to some extent correspondingly. The main degraded products and their compositions were compared (Tables S1, S2). The main carbohydrate products were glucose (43.57%, relative peak area), fructose (4.93%), and cellobiose (10.95%) at 120 °C. The glucose-degraded products of pyruvaldehyde and gluconic acid were also detected, and the compositions were relatively much low. There was still a large unknown peak in the spectrum, and it might be due to the limitations of the analytical column and chromatographic conditions, making it impossible to analyze all the components in the liquid product. However, the products in hydrolysates changed greatly with the increasing of temperature 140 °C. About 29.29% of glucose was converted to gluconic acid under the strong oxidative power of $FeCl_3$ [46–49]. The yield of pyruvaldehyde also increased from 2.46% to 5.42%. Besides, polar organic products such as ketones, aldehydes and acids were detected by GC-MS. The 5-HMF and furfural accounted for the most part of hydrolysis products. With the increasing temperature, they were further degraded to formic acid and levulinic acid.

*3.3.3. Effect of time on cellulose conversion*

The effect of reaction time on the conversion of cellulose was investigated in Fig. 9. When the reaction was carried out for 15 min, the TRS yield was only 10.6%. With the extension of the reaction time, the reducing sugar yields firstly increased and then decreased slowly with the lasting time. When the conversion time was 120 min, the TRS yield in the solution reached a maximum (66.7%), which was higher than that of previous reference [50]. After that, the yield of reducing sugar decreased to some extent, which may be due to the occurrence of the degradation reaction of monosaccharide and the formation of by-products.

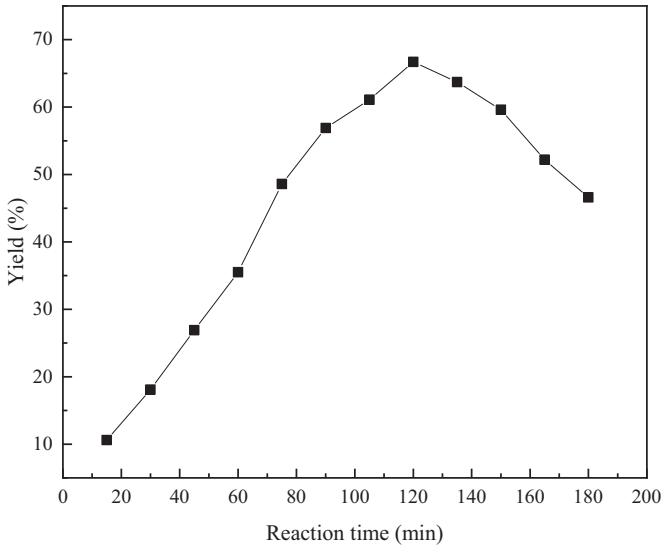



**Fig. 9.** The effect of reaction time on product yield in the conversion of cellulose.

*3.3.4. Effect of $FeCl_36H_2O$ addition on cellulose conversion*

Based on the above analysis, $FeCl_36H_2O$ was used as ligands catalyst for L-proline/ethylene glycol in the conversion of cellulose. The effect of $FeCl_36H_2O$ amount on the yield of TRS was examined (Fig. 10). The yield of TRS was increased from 35.5% to 66.7% when the molar ratio (L-proline/ethylene glycol to $FeCl_36H_2O$) increased from 0.2 to 1. However, it showed a downward with the increasing of $FeCl_36H_2O$ content from 1 to 2. The excessive $FeCl_36H_2O$ would cause the glucose continue to degrade. The results were the same with the previous studies [49]. The gluconic acid, furfural, 5-HMF, and formic acid, even levulinic acid

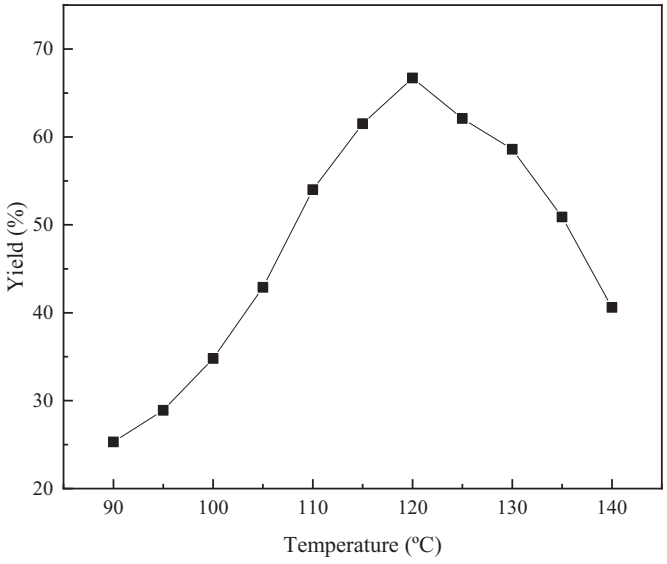



**Fig. 8.** The effect of reaction temperature on product yield in the conversion of cellulose.

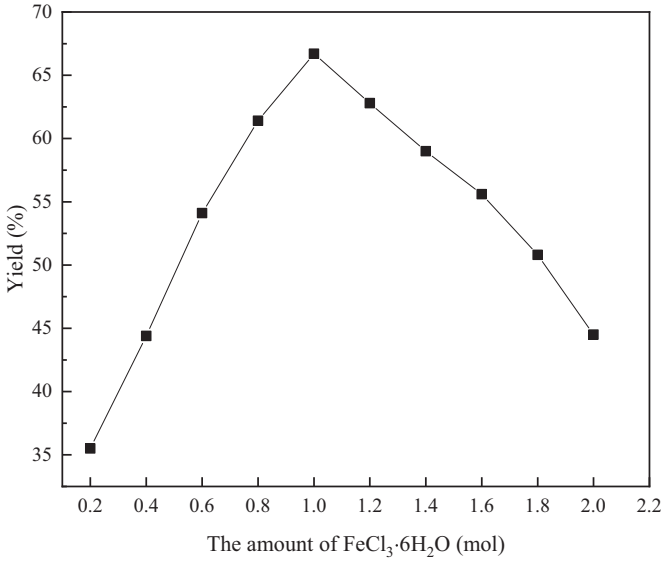



**Fig. 10.** The effect of amount of $FeCl_36H_2O$ on product yield in the conversion of cellulose.

were all detected when the $FeCl_36H_2O$ was excessive. The surface of original cellulose is neatly ordered, and most of cellulose has the crystalline structure in the SEM micrographs (Fig. S15a). However, the surface of residual cellulose solid is completely changed. The strip-like crystal cellulose become rough and disordered, and even irregular holes appeared, indicating the crystalline structure is converted to amorphous carbon (Fig. S15b). To compare the FTIR results of original cellulose and residual cellulose, there is no new peak occurred, but only slight blue shifts and enhanced peak strength. It demonstrates that the residues of cellulose hydrolysis are still dominated by cellulose (Fig. S16). In Fig. S17, the first decomposition temperatures of the two materials are 276.3 °C and 292.5 °C, respectively. The maximum weightlessness temperatures are 337.3 °C and 370.1 °C, respectively. The results meant that the thermodynamic stability of residual solid is poor than that of the original cellulose. It also can be found that the amount of residue (about 30%) generated after thermal cracking of the residue is significantly higher than that of pure cellulose (less than 17%), it might be due to the some coke and ash generated during the cellulose hydrolysis.

On the basis of experimental results and the previous literature [33], the possible reaction mechanism was proposed as Scheme 1. Firstly, raw materials L-Proline, EG, and the ligands catalyst of $FeCl_36H_2O$ reacted with each other to form O---H···N and O---H···$FeCl_36H_2O$, and the L-Pro/EG/$FeCl_36H_2O$ system was synthesized successfully. Secondly, 1,4-$\beta$ glycoside bonds in cellulose were broken greenly under the strong acidity of L-Pro/EG/$FeCl_36H_2O$. Cellulose could be transformed into total reducing sugars in the L-Pro/EG/$FeCl_36H_2O$ system.

### 3.3.5. Regeneration of NADESs

In the regeneration of NADESs, the unreacted cellulose solid and the residual reducing sugars are all need to be removed. For the unreacted cellulose solid, it could be filtered when the reaction stopped. For the most part of NADESs, the sugar is insoluble in ethanol as studied [36–40], so the reducing sugars could be precipitated from ethylene glycol and glycerol with ethanol. Because the structure of the reducing sugars are similar to ethylene glycol and glycerol, the reducing sugars would react with amino acids (L-Proline) owing to the hydrogen bond between the two materials. The residual reducing sugars also need purity from the amino acids (L-Proline). To solve this problem, the different polarity of L-Proline and the reducing sugars can be used [51]. L-Proline is a kind of non-polar amino acid, and the reducing sugars are polar substance. Therefore, the low-polarity butanone was introduced to "extract" L-Proline from the reducing sugars, and the high-purified NADES and catalyst mixture were regenerated. The reusability of NADESs was carried out, and the results showed that the TRS yield was 59.5% after eight recycles, which was slight decreased than that of the first run (66.7%) (Fig. S15). The NADESs has good catalytic stability in the hydrolysis of cellulose to TRS.

## 4. Conclusions

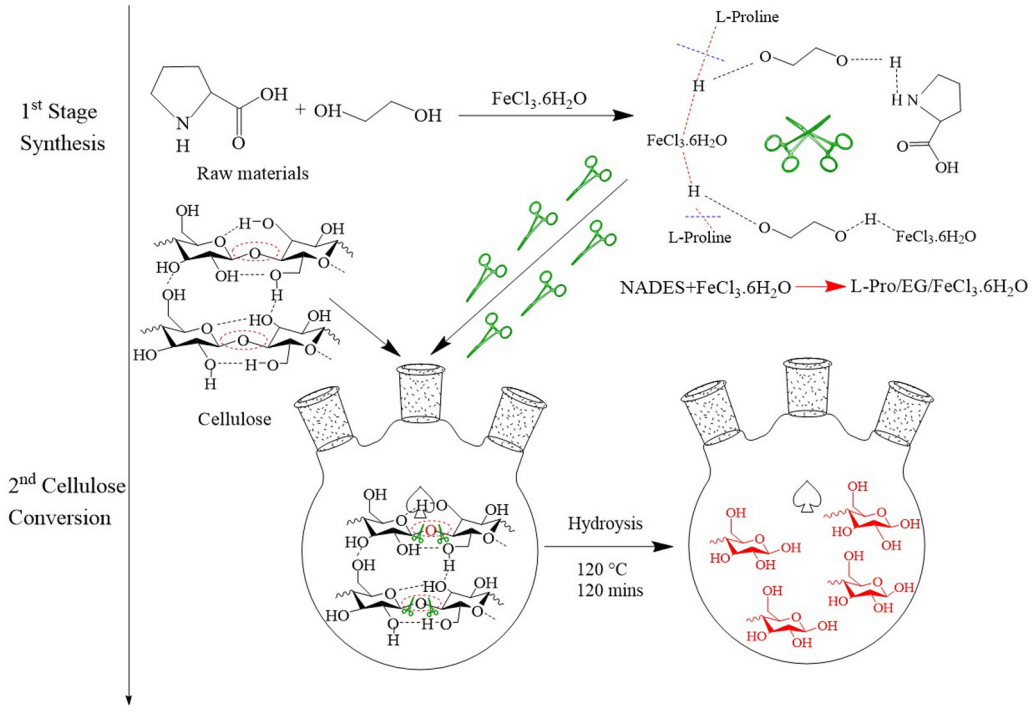
In this work, a novel series of NADESs with $FeCl_36H_2O$ as ligands catalyst were synthesized and applied in catalytic conversion of cellulose to TRS, which had great thermal stability. In the application of cellulose conversion to TRS catalyzed by NADESs, the influencing factors including the types of NADESs, reaction temperature, reaction time, and $FeCl_36H_2O$ addition amount on the TRS yield were studied. Among them, the L-Pro/EG/$FeCl_36H_2O$ system showed the best performance with 85.6% cellulose conversion and 66.7%TRS yield at reaction time 120 min and reaction temperature 120 °C. The properties of NADESs can make the processes greener, which has great potential in industrial applications.

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**Scheme 1.** Propose mechanism of L-Pro/EG/$FeCl_36H_2O$ system for converting cellulose.

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## Declaration of competing interest

No.

All authors have read the manuscript and agreed upon its submission to your journal. There are no conflicts of interest to other institutions or other people.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2020.113282.

## References

[1] US Environmental Protection Agency, Renewable Fuel Standard (RFS), http://www.epa.gov/otaq/fuels/renewablefuels/index.htm 2011.
[2] J.L. Song, H.L. Fan, J. Ma, B.X. Han, Conversion of glucose and cellulose into value-added products in water and ionic liquids, Green Chem. 15 (2013) 2619–2635.
[3] F.M.A. Geilen, B. Engendahl, A. Harwardt, W. Marquardt, J. Klankermayer, W. Leitner, Selective and flexible transformation of biomass-derived platform chemicals by a multifunctional catalytic system, Angew. Chem. Int. Ed. 49 (2010) 5510–5514.
[4] M. Fache, B. Boutevin, S. Caillol, Vanillin production from lignin and its use as a renewable chemical, ACS Sustain. Chem. Eng. 4 (2016) 35–46.
[5] X.J. Shen, J.L. Wen, P.L. Huang, K. Zheng, S.F. Wang, Q.Y. Liu, A. Charlton, R.C. Sun, Efficient and product-controlled depolymerization of lignin oriented by Raney Ni cooperated with $Cs_xH_{3-x}PW_{12}O_{40}$, BioEnergy Res. 10 (2017) 1155–1162.
[6] M. Benoit, A. Rodrigues, Q. Zhang, E. Fourre, O. Vigier Kde, J.M. Tatibouet, F. Jerome, Depolymerization of cellulose assisted by a nonthermal atmospheric plasma, Angew. Chem. Int. Ed. 50 (38) (2011) 8964–8967.
[7] L. Hu, L. Lin, Z. Wu, S. Zhou, S. Liu, Chemocatalytic hydrolysis of cellulose into glucose over solid acid catalysts, Appl. Catal. B Environ. 174 (2015) 225–243.
[8] F. Yu, R. Zhong, H. Chong, M. Smet, W. Dehaen, B.F. Sels, Fast catalytic conversion of recalcitrant cellulose into alkyl levulinates and levulinic acid in the presence of soluble and recoverable sulfonated hyperbranched poly(arylene oxindole)s, Green Chem. 9 (1) (2017) 153–163.
[9] X. Wu, X. Fan, S. Xie, J. Lin, J. Cheng, Q. Zhang, L. Chen, Y. Wang, Solar energy-driven lignin-first approach to full utilization of lignocellulosic biomass under mild conditions, Nat Catal. 1 (2018) 772–780.
[10] D.W. Wakerley, M.F. Kuehnel, K.L. Orchard, K.H. Ly, T.E. Rosser, E. Reisner, Solar-driven reforming of lignocellulose to $H_2$ with a CdS/CdOx photocatalyst, Nat. Energy 2 (2017) 17021.
[11] R.M. Wahlström, A. Suurnäkki, Enzymatic hydrolysis of lignocellulosic polysaccharides in the presence of ionic liquids, Green Chem. 17 (2) (2015) 694–714.
[12] H. Jørgensen, J.B. Kristensen, C. Felby, Enzymatic conversion of lignocellulose into fermentable sugars: challenges and opportunities, Biofuels Bioprod. Biorefin. 1 (2) (2007) 119–134.
[13] A.C. O'sullivan, Cellulose: the structure slowly unravels, Cellulose 4 (3) (1997) 173–207.
[14] P.K. Rout, A.D. Nannaware, O. Prakash, A. Kalra, R. Rajasekharan, Synthesis of hydroxymethylfurfural from cellulose using green processes: a promising biochemical and biofuel feedstock, Chem. Eng. Sci. 142 (2016) 318–346.
[15] R.P. Chandra, R. Bura, W.E. Mabee, D.A. Berlin, X. Pan, J.N. Saddler, Substrate Pretreatment: The Key to Effective Enzymatic Hydrolysis of Lignocellulosics in Biofuels, Springer, Berlin Heidelberg, 2007 67–93.
[16] Y. Nishiyama, P. Langan, H. Chanzy, Crystal structure and hydrogen-bonding system in cellulose Ip from synchrotron X-ray and neutron fiber diffraction, J. Am. Chem. Soc. 124 (31) (2002) 9074–9082.
[17] S.H. Lee, T.V. Doherty, R.J. Linhardt, J.S. Dordick, Ionic liquid-mediated selective extraction of lignin from wood leading to enhanced enzymatic cellulose hydrolysis, Biotechnol. Bioeng. 102 (5) (2009) 1368–1376.
[18] Q. Yuan, A.X. Yin, C. Luo, L.D. Sun, Y.W. Zhang, W.T. Duan, H.C. Liu, C.H. Yan, Facile synthesis for ordered mesoporous γ-aluminas with high thermal stability, J. Am. Chem. Soc. 130 (2008) 3465–3472.
[19] H. Zhang, N. Toshima, Glucose oxidation using Au-containing bimetallic and trimetallic nanoparticles, Catal. Sci. Technol. 3 (2013) 268–278.
[20] J. Wang, J. Ren, X. Liu, J. Xi, Q. Xia, Y. Zu, G. Lu, Y. Wang, Direct conversion of carbohydrates to 5-hydroxymethylfurfural using Sn-Mont catalyst, Green Chem. 14 (2012) 2506–2512.
[21] M.S. Holm, S. Saravanamurugan, E. Taarning, Conversion of sugars to lactic acid derivatives using heterogeneous zeotype catalysts, Science 328 (2010) 602–605.
[22] J.P. Lange, R. Price, P.M. Ayoub, J. Louis, L. Petrus, L. Clarke, H. Gosselink, Valeric biofuels: a platform of cellulosic transportation fuels, Angew. Chem. Int. Ed. 49 (2010) 4479–4483.
[23] Y. Román-Leshkov, C.J. Barrett, Z.Y. Liu, J.A. Dumesic, Production of dimethylfuran for liquid fuels from biomass-derived carbohydrates, Nature 447 (2007) 982–985.
[24] R. Weingarten, Y.T. Kim, G.A. Tompsett, A. Fernández, K.S. Han, E.W. Hagaman, W.C. Conner, J.A. Dumesic, G.W. Huber, Conversion of glucose into levulinic acid with solid metal (IV) phosphate catalysts, J. Catal. 304 (2013) 123–134.
[25] H. Ghaedi, M. Ayoub, S. Sufian, B. Lal, A.M. Shariff, Measurement and correlation of physicochemical properties of phosphonium-based deep eutectic solvents at several temperatures (293.15 K to 343.15 K) for $CO_2$ capture, A.M. 113 (2017) 41–51.
[26] S. Saher, H. Saleem, A.M. Asim, M. Uroos, N. Muhammad, Pyridinium based ionic liquid: a pretreatment solvent and reaction medium for catalytic conversion of cellulose to total reducing sugars (TRS), J. Mol. Liq. 272 (2018) 330–336.
[27] J.D. Mota-Morales, R.J. Sánchez-Leija, A. Carranza, J.A. Pojman, F. del Monte, G. Luna-Bárcenas, Free-radical polymerizations of and in deep eutectic solvents: green synthesis of functional materials, Prog. Polym. Sci. 78 (2017) 1139–1153.
[28] A.P. Abbott, G. Capper, D.L. Davies, R.K. Rasheed, V. Tambyrajah, Novel solvent properties of choline chloride/urea mixtures, Chem. Commun. 9 (2003) 70–71.
[29] K.D.O. Vigier, G. Chatel, F. Jérôme, Contribution of deep eutectic solvents for biomass processing: opportunities, challenges, and limitations, ChemCatChem. 7 (2015) 1250–1260.
[30] M. Francisco, A. van den Bruinhorst, M.C. Kroon, Low-transition-temperature mixtures (LTTMs): a new generation of designer solvents, Angew. Chem. 52 (2013) 3074–3085.
[31] A. Procentese, E. Johnson, V. Orr, A. Garruto Campanile, J.A. Wood, A. Marzocchella, L. Rehmann, Deep eutectic solvent pretreatment and subsequent saccharification of corncob, Bioresour. Technol. 192 (2015) 31–36.
[32] R. Wahlström, J. Hiltunen, M. Pitaluga de Souza Nascente Sirkka, S. Vuoti, K. Kruus, Comparison of three deep eutectic solvents and 1-ethyl-3-methylimidazolium acetate in the pretreatment of lignocellulose: effect on enzyme stability, lignocellulose digestibility and one-pot hydrolysis, RSC Adv. 6 (72) (2016) 68100–68110.
[33] H. Zhang, N. Li, X. Pan, S. Wu, J. Xie, Direct transformation of cellulose to gluconic acid in a concentrated Iron(III) chloride solution under mild conditions, ACS Sustain. Chem. Eng. 5 (2017) 4066–4072.
[34] A. Hayyan, M.A. Hashim, M. Hayyan, F.S. Mjalli, I.M. AlNashef, A new processing route for cleaner production of biodiesel fuel using a choline chloride based deep eutectic solvent, J. Clean. Prod. 65 (2014) 246–251.
[35] H. Zhang, N. Li, X. Pan, S. Wu, J. Xie, Oxidative conversion of glucose to gluconic acid by iron (III) chloride in water under mild conditions, Green Chem. 8 (2016) 2308–2312.
[36] X. Yan, Short-chain polysaccharide analysis in ethanol–water solutions, J. AOAC Int. 100 (2017) 1134–1136.
[37] B. Bello, S. Mustafa, J.S. Tan, T.A.T. Ibrahim, Y.J. Tam, A.B. Ariff, M.Y. Manap, S. Abbasiliasi, Evaluation of the effect of soluble polysaccharides of palm kernel cake as a potential prebiotic on the growth of probiotics, 3 Biotech 8 (2018) 346.
[38] S. Kassaye, K.K. Pant, S. Jain, Hydrolysis of cellulosic bamboo biomass into reducing sugars via a combined alkaline solution and ionic liquid pretreament steps, Renew. Energy 104 (2017) 177–184.
[39] B. Shi, Sucrose solubility in alcohol-water solution, Guangzhou Chemical Industry 37 (2009) 101–102.
[40] W. Liu, Y. Hou, W. Wu, S. Ren, Y. Jing, B. Zhang, Solubility of glucose in ionic liquid + antisolvent mixtures, Ind. Eng. Chem. Res. 50 (2011) 6952–6956.
[41] A.P. Abbott, G. Capper, D.L. Davies, K.R. Rasheed, V. Tambyrajah, Novel solvent properties of choline chloride/urea mixtures, Chem. Commun. (2003) 70–71.
[42] C. Florindo, F.S. Oliveira, L.P.N. Rebelo, A.M. Fernandes, I.M. Marrucho, Insights into the synthesis and properties of deep eutectic solvents based on cholinium chloride and carboxylic acid, ACS Sustainable Chemistry Engineering 2 (2014) 2416–2425.
[43] A.P. Abbott, D. Boothby, G. Capper, D.L. Davies, R.K. Rasheed, Deep eutectic solvents formed between choline chloride and carboxylic acids: versatile alternatives to ionic liquids, J. Am. Chem. Soc. 126 (2004) 9142–9147.
[44] G. García, S. Aparicio, R. Ullah, M. Atilhan, Deep eutectic solvents: physicochemical properties and gas separation applications, Energy Fuel 29 (2015) 616–2644.

[45] H. Ghaedi, M. Ayoub, S. Sufian, S.M. Hailegiorgis, G. Murshid, S.N. Khan, Thermal stability analysis, experimental conductivity and pH of phosphoniumbased deep eutectic solvents and their prediction by a new empirical equation, J. Chem. Thermodyn. 116 (2018) 50–60.
[46] H. Zhang, N. Li, X. Pan, S. Wu, J. Xie, Oxidative conversion of glucose to gluconic acid by iron (III) chloride in water under mild conditions, Green Chem. 8 (2016) 2308–2312.
[47] L. Logenius, L. Friman, R. Agnemo, The influence of temperature and moisture on the optical properties of cellulose in the presence of metal chlorides and glucuronic acid, Nordic Pulp & Paper Research Journal 23 (2008) 72–80.
[48] V.S. Šelih, M. Strlič, J. Kolar, B. Pihlar, The role of transition metals in oxidative degradation of cellulose, Polym. Degrad. Stab. 92 (2007) 1476–1481.
[49] F. Liu, Z. Xue, X. Zhao, H. Mou, J. He, T. Mu, Catalytic deep eutectic solvents for highly efficient conversion of cellulose to gluconic acid with gluconic acid self-precipitation separation, Chem. Commun. 54 (2018) 6140–6143.
[50] K. Shimizu, T. Higuchi, E. Takasugi, T. Hatamachi, T. Kodama, A. Satsuma, Characterization of Lewis acidity of cation-exchanged montmorillonite K-10 clay as effective heterogeneous catalyst for acetylation of alcohol, J. Mol. Catal. A Chem. 284 (2018) 89–96.
[51] M. Zuo, K. Le, Z. Li, Y. Jiang, X. Zeng, X. Tang, Y. Sun, L. Lin, Green process for production of 5-hydroxymethylfurfural from carbohydrates with high purity in deep eutectic solvents, Ind. Crop. Prod. 99 (2017) 1–6.